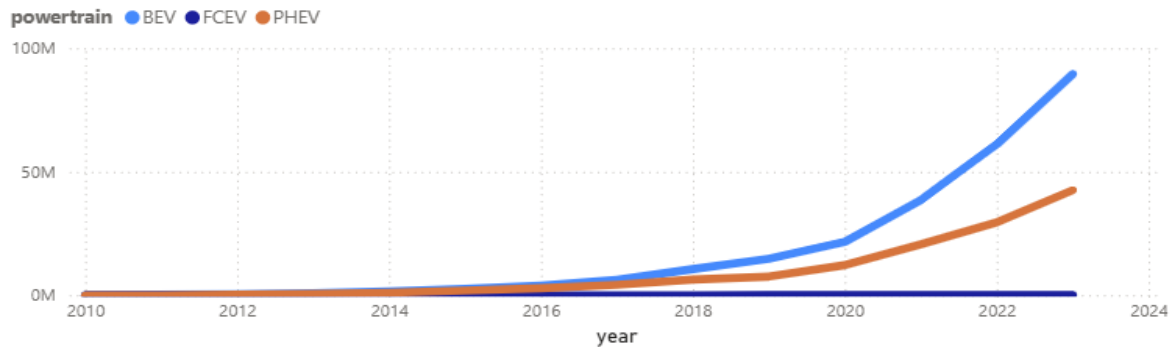


## BEV Takes the Lead – A Market in Motion

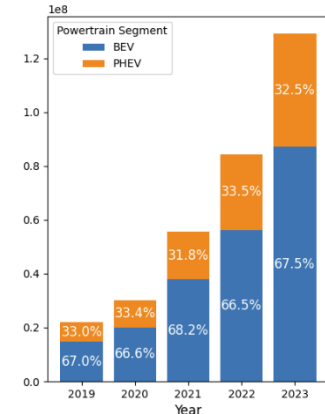
In the last ten years, all powertrain types have grown – yet BEVs have surged ahead, dominating the market in both volume and value. The line chart illustrates this dominance in *absolute sales value*, with BEV accelerating sharply in recent years.

Long-Term Sales Value Trends Across Powertrain segments



Over the past five years, BEVs have consistently held nearly 70% of the annual market share by value. While the overall market has expanded, Plug-in Hybrid Electric Vehicles (PHEVs) have gradually declined in relative terms. Fuel Cell Electric Vehicles (FCEVs) remain marginal, consistently representing less than 3% of total sales. Due to their limited market presence, they have been excluded from the bar chart to better highlight the dynamics between BEVs and PHEVs.

Market Overview - Powertrain Sales Value Over Time



## Different Metrics, Different Winners – What Are We Really Comparing?

Electric vehicles excel in different areas depending on what you choose to measure. Some dominate in range, others in acceleration, horsepower, or price. The chart below illustrates this diversity, showing how various models rise to the top depending on the metric. Lucid Motors Air leads in range and acceleration, while Chevrolet Bolt EV stands out in affordability. But these isolated comparisons raise a broader question: is there a consistent relationship between performance and price?

Top Value Picks – Price Index Leaders



Quickest Off the Line – Acceleration Index



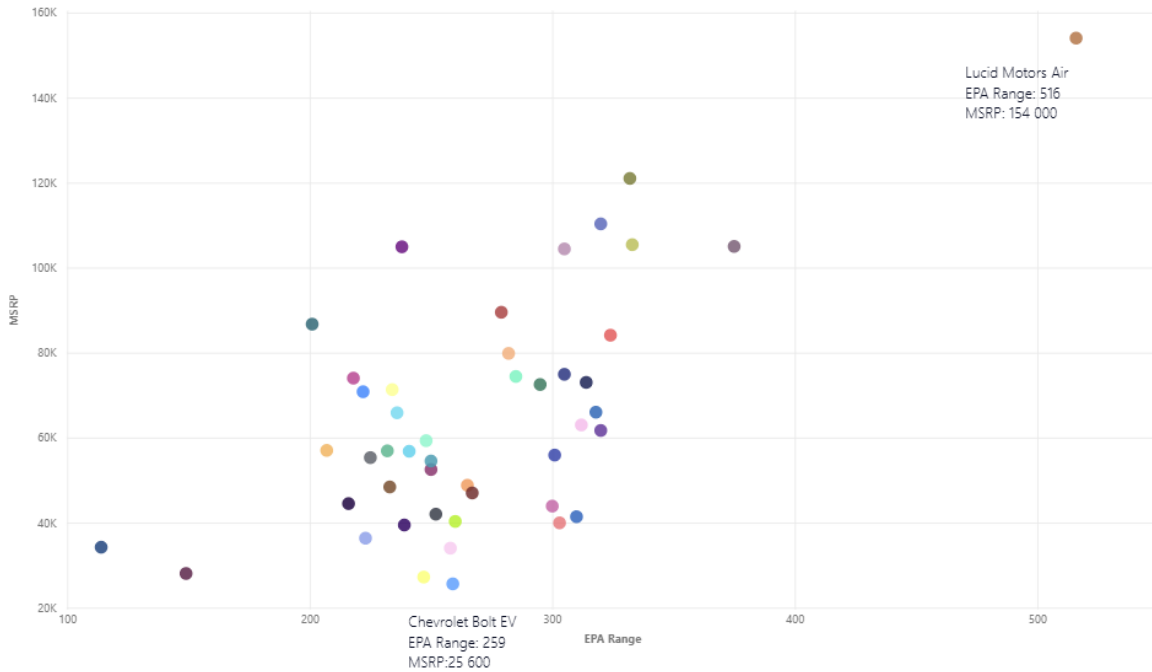
Longest Range Champions – EPA Range Index



Powerhouses on Wheels – Horsepower Index



Relationship Between EPA Range and MSRP in Electric Vehicles



When we look at range versus MSRP, the scatterplot reveals a loose correlation – longer range often comes at a higher cost – but not always. Outliers like the Hyundai Kona Electric and Chevrolet Bolt EV challenge the assumption that premium performance must mean premium pricing.

To enable a fair and balanced comparison across models, five key parameters were normalized to a 0–100 scale: acceleration, horsepower, mass, EPA range, and price. Higher scores reflect better performance or value (e.g., faster acceleration, longer range, lower price). These individual indices were then averaged into a single **Performance Index**, offering a composite view of each vehicle’s overall capability.

### "And the Winner Is..."

*When performance is distilled into a single, balanced metric – one that considers acceleration, horsepower, battery capacity, top speed, range, and price – a clear leader emerges. Among the contenders, one model consistently excels across the board, combining speed with stamina, power with precision, and value with innovation. Tesla Model S stands at the top, not just as a symbol of electric mobility, but as the most well-rounded performer in today's EV landscape. It's not merely fast or efficient – it's the benchmark against which others are measured*

## Tesla Model S

Top Performance Index

| Make/Model                             | Performance Index | Acceleration Index | EPA Range Index | Horsepower Index | Mass Index | Price Index |
|----------------------------------------|-------------------|--------------------|-----------------|------------------|------------|-------------|
| Tesla Model S                          | 68.48             | 98.00              | 64.93           | 60.38            | 80.91      | 38.17       |
| Ford Mustang Mach-E                    | 66.35             | 90.00              | 46.27           | 37.58            | 72.14      | 85.75       |
| Rivian R1T                             | 64.93             | 90.00              | 49.75           | 80.19            | 41.63      | 63.08       |
| Lucid Motors Air                       | 61.24             | 100.00             | 100.00          | 37.58            | 68.63      | 0.00        |
| Tesla Model Y                          | 60.99             | 72.00              | 50.75           | 34.57            | 79.07      | 68.54       |
| Tesla Model 3                          | 60.67             | 58.00              | 38.06           | 34.57            | 89.36      | 83.34       |
| Ford F-150 Lightning                   | 60.59             | 80.00              | 51.24           | 49.58            | 50.22      | 71.89       |
| Rivian R1S                             | 59.70             | 70.00              | 45.02           | 80.19            | 39.79      | 63.47       |
| Cadillac LYRIQ                         | 58.25             | 68.00              | 49.25           | 39.98            | 63.14      | 70.88       |
| Genesis Electrified GV70               | 58.15             | 76.00              | 30.35           | 31.45            | 84.32      | 68.65       |
| BMW iX xDrive50                        | 57.91             | 80.00              | 52.24           | 41.90            | 60.96      | 54.44       |
| Mercedes-Benz EQS Sedan                | 57.31             | 78.00              | 47.51           | 57.86            | 64.54      | 38.63       |
| Polestar 2                             | 56.62             | 66.00              | 29.60           | 28.21            | 77.05      | 82.24       |
| GMC Hummer EV                          | 55.86             | 94.00              | 51.24           | 100.00           | 0.00       | 34.04       |
| Tesla Model X                          | 55.79             | 84.00              | 54.23           | 60.38            | 54.62      | 25.71       |
| Volvo C40                              | 55.61             | 66.00              | 27.61           | 28.21            | 79.37      | 76.87       |
| Nissan Ariya                           | 54.82             | 60.00              | 25.37           | 26.65            | 76.78      | 85.29       |
| Audi e-tron GT                         | 54.63             | 98.00              | 30.85           | 36.25            | 69.79      | 38.24       |
| Mercedes-Benz EQE Sedan                | 54.39             | 40.00              | 47.51           | 54.02            | 68.80      | 61.60       |
| Genesis Electrified G80                | 54.23             | 78.00              | 41.79           | 23.77            | 69.83      | 57.77       |
| Audi e-tron Sportback                  | 54.20             | 94.00              | 25.87           | 28.21            | 60.62      | 62.31       |
| Genesis GV60                           | 54.06             | 50.00              | 33.33           | 31.45            | 81.76      | 73.76       |
| BMW i4 eDrive40                        | 53.95             | 50.00              | 46.52           | 20.17            | 76.65      | 76.40       |
| Volkswagen ID.4                        | 53.43             | 46.00              | 36.32           | 15.37            | 80.91      | 88.56       |
| Jaguar I-PACE                          | 53.06             | 70.00              | 29.85           | 27.25            | 73.82      | 64.41       |
| Chevrolet Bolt EV                      | 52.78             | 30.00              | 36.07           | 3.96             | 93.86      | 100.00      |
| Nissan LEAF                            | 51.62             | 50.00              | 8.71            | 5.64             | 95.67      | 98.10       |
| Chevrolet Bolt EUV                     | 51.60             | 30.00              | 33.08           | 3.96             | 92.19      | 98.75       |
| Porsche Taycan                         | 51.36             | 58.00              | 21.64           | 53.90            | 70.85      | 52.41       |
| Hyundai Kona Electric                  | 50.89             | 32.00              | 35.82           | 4.08             | 89.09      | 93.46       |
| VinFast VF 8                           | 50.54             | 50.00              | 23.13           | 28.21            | 75.79      | 75.55       |
| Audi Q4 e-tron                         | 50.35             | 36.00              | 37.56           | 15.37            | 80.91      | 81.93       |
| Hyundai IONIQ 5                        | 50.07             | 12.00              | 47.01           | 18.37            | 84.15      | 88.82       |
| Volvo XC40                             | 49.46             | 24.00              | 27.11           | 9.60             | 94.95      | 91.63       |
| Kia Niro EV                            | 48.39             | 26.00              | 31.09           | 4.08             | 91.54      | 89.21       |
| Mercedes-Benz EQE SUV                  | 48.22             | 34.00              | 41.04           | 54.02            | 61.81      | 50.23       |
| Toyota bZ4X                            | 47.83             | 30.00              | 34.33           | 5.64             | 81.93      | 87.23       |
| Audi Q8 e-tron                         | 46.86             | 44.00              | 42.54           | 22.57            | 63.18      | 61.99       |
| Audi Q4 e-tron Sportback               | 46.82             | 36.00              | 31.59           | 15.37            | 75.45      | 75.70       |
| Lordstown Endurance                    | 46.42             | 34.00              | 33.83           | 32.77            | 52.44      | 79.05       |
| Audi e-tron                            | 45.99             | 50.00              | 26.87           | 22.57            | 65.73      | 64.80       |
| Kia EV6                                | 45.99             | 16.00              | 48.76           | 0.00             | 77.50      | 87.69       |
| Mercedes-Benz EQB                      | 44.19             | 20.00              | 33.83           | 14.53            | 75.11      | 77.49       |
| Mini Cooper Electric SE Hardtop 2 door | 41.79             | 14.00              | 0.00            | 1.68             | 100.00     | 93.28       |
| Mercedes-Benz EQS SUV                  | 39.64             | 30.00              | 54.48           | 22.57            | 53.29      | 37.85       |
| Mercedes-Benz EQB SUV                  | 38.92             | 0.00               | 29.35           | 14.53            | 75.11      | 75.62       |

## ***Methodology and Evaluation Framework***

### **Methodology and Tools**

The analysis is based on data from *Cars\_individuell\_tentamen.xlsx* (Cars2 sheet) and *Cars\_country\_individuell\_tentamen.csv*. Battery Electric Vehicles (BEVs) were isolated for focused comparison.

All data cleaning, normalization, and index calculations were performed in Python using Jupyter Notebook, ensuring reproducibility and transparency. Five performance-related parameters were scaled into indices (0–100), then averaged into a composite Performance Index.

Visualizations were created in Power BI, as required, and manually inserted into the Word document to meet formatting guidelines. Supporting datasets were split into volume, share, and energy files to enable targeted analysis.

### **Visualization and Comparison**

The table below presents all BEV models along with their respective index values. It serves as the foundation for both the bar and scatterplot visualizations in this report, clearly illustrating the diversity in performance and pricing across the EV landscape.

### **Conclusion**

The analysis identifies the **Tesla Model S** as the most well-rounded performer, based on the composite Performance Index. It combines speed, range, power, and value in a way that sets the benchmark for today's electric vehicles.

Given BEVs' rapid growth and market dominance, this report has focused exclusively on this powertrain category. The findings reinforce the notion that BEVs are not just the future of mobility – they are already leading the way.